

Exercises for Process Mining

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Notes Repository

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1 Process Discovery

1.1 Alpha Miner

1.1.1 Exercise 1

Given the following event log:

$$L_1 = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^2, \langle a, e, d \rangle^1]$$

build the process model using the Alpha Miner algorithm.

Solution

1.

1.1.2 Exercise 2

Given the following event log:

$$L = [\langle a, b, c, d, e, f, b, d, c, e, g \rangle, \langle a, b, d, c, e, g \rangle^2, \langle a, b, c, d, e, f, b, c, d, e, f, b, d, c, e, g \rangle]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1. First, we need to compute the footprint matrix:

	A	B	C	D	E	F	G
A	#	→	#	#	#	#	#
B		#	→	→	#	←	#
C			#		→	#	#
D				#	→	#	#
E					#	→	→
F						#	#
G							#

Table 1: Footprint Matrix for the Event Log L

2. Next, we identify the sets of all transitions T_L , the set of start activities T_I and the set of end activities T_O :

- $T_L = \{a, b, c, d, e, f, g\}$
- $T_I = \{a\}$
- $T_O = \{g\}$

3. Then, we identify the pairs of sets of transitions that satisfy the Alpha Miner conditions. To find the first pairs, we look for causality relations in the footprint matrix:

$$\{(\{a\}, \{b\}), (\{b\}, \{c\}), (\{b\}, \{d\}), (\{f\}, \{b\}), (\{c\}, \{e\}), (\{d\}, \{e\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

4. We can then group these pairs into maximal sets:

$$\{(\{a, f\}, \{b\}), (\{e\}, \{f, g\}), \}$$

And we can remove the non-maximal sets that composed them:

$$\{(\{a\}, \{b\}), (\{f\}, \{b\}), (\{e\}, \{f\}), (\{e\}, \{g\})\}$$

5. Finally, we can construct the Workflow net, using one place for each of the identified pairs, and connecting them according to the causality relations:

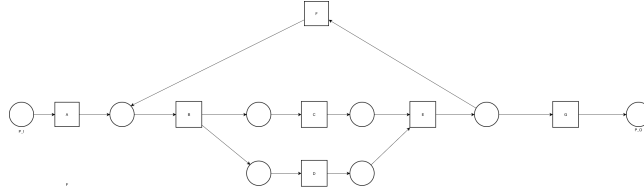


Figure 1: Petri Net Discovered using the Alpha Miner

1.1.3 Exercise 3

Given the following event log:

$$L_4 = [\langle a, c, d \rangle^{45}, \langle b, c, d \rangle^{42}, \langle a, c, e \rangle^{38}, \langle b, c, e \rangle^{22}]$$

build the process model using the Alpha Miner algorithm.

Solution

- 1.

1.1.4 Exercise 4

Consider the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Use the Alpha Miner algorithm to discover a Petri net from the event log.

Solution:

1.2 Heuristic Miner

1.2.1 Exercise 1

Given the following event log:

$$L = [\langle a, e \rangle^5, \langle a, b, c, e \rangle^{10}, \langle a, c, b, e \rangle^{10}, \langle a, b, e \rangle^1, \langle a, c, e \rangle^1, \\ \langle a, d, e \rangle^{10}, \langle a, d, d, e \rangle^2, \langle a, d, d, d, e \rangle^1]$$

use the Heuristic Miner algorithm to discover a process model from the event log, with a threshold of 0.7:

Solution:

1. First, we need to compute the frequency matrix:

	A	B	C	D	E
A	0	11	11	13	5
B	0	0	10	0	11
C	0	10	0	0	11
D	0	0	0	4	13
E	0	0	0	0	0

Table 2: Frequency Matrix for the Event Log L

Each cell (x, y) contains the number of times activity x is directly followed by activity y in the event log.

2. Next, we compute the dependency matrix, by applying the dependency measure formula to each pair of activities. Remember that the dependency measure is defined as:

$$\begin{cases} \frac{|x \rightarrow y| - |y \rightarrow x|}{|x \rightarrow y| + |y \rightarrow x| + 1} & \text{if } x \neq y \\ \frac{|x \rightarrow x|}{|x \rightarrow x| + 1} & \text{otherwise} \end{cases}$$

Note that:

- Along the diagonal, we apply the second case of the formula;
- We can compute the upper triangle of the matrix, the lower triangle can be derived from it by simply swapping signs.

	A	B	C	D	E
A	0	$\frac{11}{12}$	$\frac{11}{12}$	$\frac{13}{14}$	$\frac{5}{14}$
B		0	$\frac{10-10}{21}$	0	$\frac{11}{12}$
C			0	0	$\frac{11}{12}$
D				$\frac{4}{5}$	$\frac{13}{14}$
E					0

Table 3: Dependency Matrix for the Event Log L

3. We can then build the dependency graph, using the given threshold. We create a node for each activity in the log, and add the directed edges for each pair of activities whose dependency measure is above the threshold:

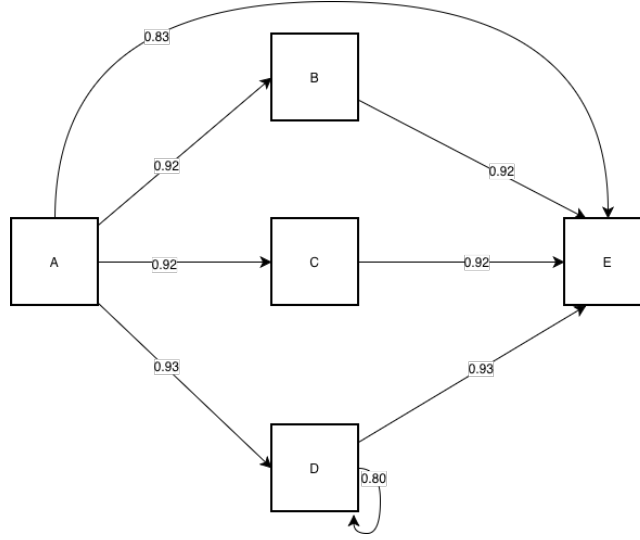


Figure 2: Dependency Graph for the Event Log L

1.2.2 Exercise 2

Given the following event log:

$$L_2 = [\langle a, b, d, e \rangle^{80}, \langle a, d, b, e \rangle^{30}, \langle a, e \rangle^2, \langle a, b, c, d, e \rangle^{10}]$$

use the Heuristic Miner algorithm to discover the dependency matrix from the event log and build the dependency graph with a threshold of 0.9.

1.2.3 Exercise 3

Given the following event log:

$$L_3 = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Compute:

- The frequency matrix;
- The dependency matrix;
- The dependency graph with a threshold of 0.9 and a threshold for direct succession of 1.

1.3 Region Based Miner

1.3.1 Exercise 1

Construct a transition system applying the *past with multiset abstraction* for the given log:

$$L = [\langle a, c, b, d \rangle, \langle a, b, c, d \rangle, \langle b, a, c, d \rangle]$$

Solution:

1. We create a starting state q_0 with an empty multiset;
2. We replay each trace in the log, updating the multiset and creating new states as needed. For instance, considering the first trace $\langle a, c, b, d \rangle$:
 - For the event a , we add a transition from q_0 to a new state q_1 with multiset $\{a\}$;
 - For the event c , we add a transition from q_1 to q_2 with multiset $\{a, c\}$;
 - For the event b , we add a transition from q_2 to q_3 with multiset $\{a, b, c\}$;
 - ...

Continuing this process for all traces, we obtain the following transition system:

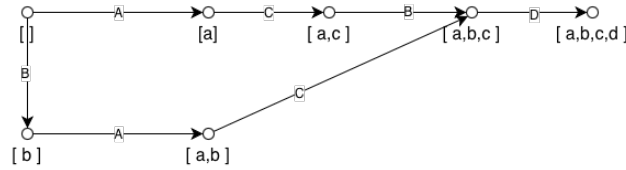


Figure 3: Transition System using Past with Multiset Abstraction

1.3.2 Exercise 2

Given the previous transition system, compute the resulting Petri net:

Solution:

1. We identify the minimal, non-trivial regions in the transition system. In this case, we have:

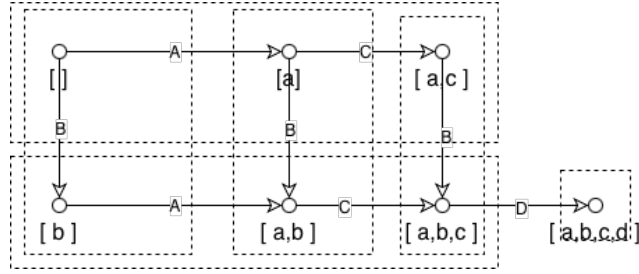


Figure 4: Identified Regions from the Transition System

2. We then create a place for each region. All the transition entering a region will have an arc to the corresponding place, and all transitions leaving the region will have an arc from the place. This results in the following Petri net:

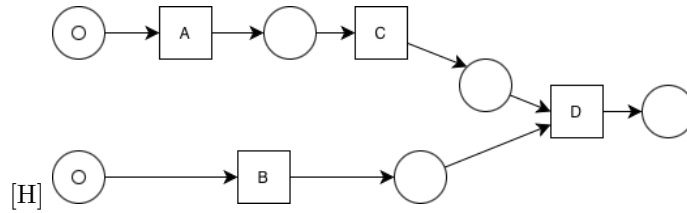


Figure 5: Petri Net Derived from the Transition System

1.3.3 Exercise 3

Given the following event log:

$$L = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^2, \langle a, e, d \rangle^1]$$

build the process model using the Region Based Miner algorithm with *past with set abstraction*.

Solution:

1.3.4 Exercise 4

Given the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

Use the Region Based Miner algorithm with *past with set abstraction* to discover a Petri Net:

1.4 Inductive Miner

1.4.1 Exercise 1

Given the following event log:

$$L = [\langle a, b, c, d \rangle^3, \langle a, c, b, d \rangle^4, \langle a, b, c, e, f, b, c, d \rangle^2, \langle a, c, b, e, f, b, c, d \rangle^2, \\ \langle a, b, c, e, f, c, b, d \rangle, \langle a, c, b, e, f, b, c, e, f, c, b, d \rangle]$$

use the Inductive Miner algorithm to discover a process model from the event log.

Solution:

1.4.2 Exercise 2

Given the following event log:

$$L = [\langle a, c, d \rangle^{45}, \langle b, c, d \rangle^{42}, \langle a, c, e \rangle^{38}, \langle b, c, e \rangle^{22}]$$

use the Inductive Miner algorithm to discover a process model from the event log.

1.4.3 Exercise 3

Given the following event log:

$$L = [\langle a, b, d \rangle^{100}, \langle a, c, b, d \rangle^{80}, \langle a, b, c, d \rangle^2, \langle a, c, d \rangle^{40}, \langle a, d \rangle^1]$$

use the Inductive Miner algorithm to discover a process model from the event log.